

# Module 4: TDD using JUnit 5 and Mockito & SLF4J

## Logging Framework

### 1. Setting up JUnit

The Project Directory is as follows:

untitled [JUnitDemo]

```
|—.idea
|—src
|   |—.main
|   |   |—.java
|   |   |—.org
|   |   |—.example
|   |   |—.App.java
|   |—.test
|   |—.java
|   |—.org
|   |—.example
|   |—.AssertionsTest.java
|—target
|—.gitignore
|—pom.xml
```

#### Codes:

##### CalculatorTest.java

```
import org.junit.Test;
import static org.junit.Assert.*;
public class CalculatorTest {
    @Test
    public void testAddition() {
        int result = 2 + 3;
        assertEquals(5, result);
    }
}
```

##### pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
http://maven.apache.org/xsd/maven-4.0.0.xsd">
    <modelVersion>4.0.0</modelVersion>
    <groupId>org.example</groupId>
    <artifactId>JUnitDemo</artifactId>
    <version>1.0-SNAPSHOT</version>
    <packaging>jar</packaging>
    <name>untitled</name>
    <url>http://maven.apache.org</url>
    <properties>
        <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
```

```

</properties>
<dependencies>
  <dependency>
    <groupId>junit</groupId>
    <artifactId>junit</artifactId>
    <version>4.13.2</version>
    <scope>test</scope>
  </dependency>
</dependencies>
</project>

```

## 2. Assertions in JUnit

The Project Directory is as follows:

untitled [JUnitDemo]

```

├── .idea
├── src
│   ├── main
│   │   ├── java
│   │   ├── org
│   │   │   ├── example
│   │   │   └── App.java
│   └── test
│       ├── java
│       ├── org
│       └── example
│           └── AssertionsTest.java
├── target
├── .gitignore
└── pom.xml

```

### Codes:

#### AssertionsTest.java

```

package org.example;
import org.junit.Test;
import static org.junit.Assert.*;
public class AssertionsTest {
    @Test
    public void testAssertions() {
        // Assert equals
        assertEquals(5, 2 + 3);
        // Assert true
        assertTrue(5 > 3);
        // Assert false
        assertFalse(5 < 3);
        // Assert null
        assertNull(null);
        // Assert not null
        assertNotNull(new Object());
    }
}

```

## pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
http://maven.apache.org/xsd/maven-4.0.0.xsd">
  <modelVersion>4.0.0</modelVersion>
  <groupId>org.example</groupId>
  <artifactId>JUnitDemo</artifactId>
  <version>1.0-SNAPSHOT</version>
  <packaging>jar</packaging>
  <name>untitled</name>
  <url>http://maven.apache.org</url>
  <properties>
    <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
  </properties>
  <dependencies>
    <dependency>
      <groupId>junit</groupId>
      <artifactId>junit</artifactId>
      <version>4.13.2</version>
      <scope>test</scope>
    </dependency>
  </dependencies>
</project>
```

### 3. Arrange-Act-Assert (AAA) Pattern, Test Fixtures, Setup and Teardown Methods in Junit

The full Project Directory is as follows:

untitled [JUnitDemo]

```
├── .idea
├── src
│   ├── main
│   │   ├── java
│   │   ├── org
│   │   └── example
│   │       ├── App.java
│   │       └── Calculator.java
│   └── test
│       ├── java
│       ├── org
│       └── example
│           ├── AAATest.java
│           └── AssertionsTest.java
├── target
├── .gitignore
└── pom.xml
```

## Codes:

### Calculator.java

```
package org.example;
public class Calculator {
    public int add(int a, int b) {
        return a + b;
    }
    public int subtract(int a, int b) {
        return a - b;
    }
    public int multiply(int a, int b) {
        return a * b;
    }
}
```

### AAATest.java

```
package org.example;
import org.junit.After;
import org.junit.Before;
import org.junit.Test;
import static org.junit.Assert.*;
public class AAATest {
    private Calculator calculator;
    // Setup Method
    @Before
    public void setUp() {
        System.out.println("Setting up Calculator...");
        calculator = new Calculator();
    }
    // Test Method using AAA Pattern
    @Test
    public void testAddition() {
        // Arrange
        int a = 10;
        int b = 5;
        // Act
        int result = calculator.add(a, b);
        // Assert
        assertEquals(15, result);
    }
    // Another Test
    @Test
    public void testSubtraction() {
        // Arrange
        int a = 20;
        int b = 8;
        // Act
        int result = calculator.subtract(a, b);
        // Assert
        assertEquals(12, result);
    }
    // Teardown Method
    @After
    public void tearDown() {
        System.out.println("Cleaning up...");
        calculator = null;
    }
}
```

## pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
http://maven.apache.org/xsd/maven-4.0.0.xsd">
  <modelVersion>4.0.0</modelVersion>
  <groupId>org.example</groupId>
  <artifactId>JUnitDemo</artifactId>
  <version>1.0-SNAPSHOT</version>
  <packaging>jar</packaging>
  <name>untitled</name>
  <url>http://maven.apache.org</url>
  <properties>
    <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
  </properties>
  <dependencies>
    <dependency>
      <groupId>junit</groupId>
      <artifactId>junit</artifactId>
      <version>4.13.2</version>
      <scope>test</scope>
    </dependency>
  </dependencies>
</project>
```

## 4. Mocking and Stubbing

The full Project Directory is as follows:

MockitoDemo

```
|— .idea
|— src
|   |— main
|   |   |— java
|   |   |   |— com
|   |   |       |— example
|   |   |       |— ExternalApi.java
|   |   |       |— MyService.java
|   |   |       |— resources
|   |— test
|   |   |— java
|   |   |   |— com
|   |   |       |— example
|   |   |       |— MyServiceTest.java
|— target
|— .gitignore
|— pom.xml
```

### Codes:

#### ExternalApi.java

```
package com.example;
```

```
public interface ExternalApi {  
    String getData();  
}
```

### MyService.java

```
package com.example;  
public class MyService {  
    private ExternalApi api;  
    public MyService(ExternalApi api) {  
        this.api = api;  
    }  
    public String fetchData() {  
        return api.getData();  
    }  
}
```

### MyServiceTest.java

```
package com.example;  
import org.junit.jupiter.api.Test;  
import static org.junit.jupiter.api.Assertions.assertEquals;  
import static org.mockito.Mockito.*;  
public class MyServiceTest {  
    @Test  
    public void testExternalApi() {  
        // Step 1: Create mock object  
        ExternalApi mockApi = mock(ExternalApi.class);  
        // Step 2: Stub method  
        when(mockApi.getData()).thenReturn("Mock Data");  
        // Step 3: Inject mock into service  
        MyService service = new MyService(mockApi);  
        // Step 4: Call method  
        String result = service.fetchData();  
        // Step 5: Verify result  
        assertEquals("Mock Data", result);  
    }  
}
```

### pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"  
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"  
    xsi:schemaLocation="http://maven.apache.org/POM/4.0.0  
        https://maven.apache.org/xsd/maven-4.0.0.xsd">  
    <modelVersion>4.0.0</modelVersion>  
    <groupId>com.example</groupId>  
    <artifactId>MockitoDemo</artifactId>  
    <version>1.0-SNAPSHOT</version>  
    <properties>  
        <maven.compiler.source>17</maven.compiler.source>  
        <maven.compiler.target>17</maven.compiler.target>  
        <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>  
    </properties>  
    <dependencies>  
        <!-- JUnit 5 -->  
        <dependency>  
            <groupId>org.junit.jupiter</groupId>  
            <artifactId>junit-jupiter</artifactId>  
            <version>5.10.2</version>  
            <scope>test</scope>  
        </dependency>  
        <!-- Mockito -->  
        <dependency>
```

```

        <groupId>org.mockito</groupId>
        <artifactId>mockito-core</artifactId>
        <version>5.12.0</version>
        <scope>test</scope>
    </dependency>
</dependencies>
<build>
    <plugins>
        <plugin>
            <groupId>org.apache.maven.plugins</groupId>
            <artifactId>maven-surefire-plugin</artifactId>
            <version>3.2.5</version>
        </plugin>
    </plugins>
</build>
</project>

```

## 5. Verifying Interactions

The full Project Directory is as follows:

MockitoVerifyDemo

```

├── .idea
├── src
│   ├── main
│   │   ├── java
│   │   │   ├── com
│   │   │   │   ├── example
│   │   │   │   │   ├── ExternalApi.java
│   │   │   │   │   ├── MyService.java
│   │   │   │   │   └── resources
│   │   └── test
│   │       ├── java
│   │       │   ├── com
│   │       │   │   ├── example
│   │       │   │   │   └── MyServiceTest.java
│   └── target
├── .gitignore
└── pom.xml

```

### Codes:

#### ExternalApi.java

```

package com.example;
public interface ExternalApi {
    String getData();
}

```

#### MyService.java

```

package com.example;
public class MyService {
    private ExternalApi api;
    public MyService(ExternalApi api) {

```

```

        this.api = api;    }
    public String fetchData() {
        return api.getData();
    }
}

```

## MyServiceTest.java

```

package com.example;
import org.junit.jupiter.api.Test;
import static org.mockito.Mockito.*;
public class MyServiceTest {
    @Test
    public void testVerifyInteraction() {
        // Step 1: Create mock object
        ExternalApi mockApi = mock(ExternalApi.class);
        // Step 2: Inject mock into service
        MyService service = new MyService(mockApi);
        // Step 3: Call the service method
        service.fetchData();
        // Step 4: Verify that getData() was called
        verify(mockApi).getData();
    }
}

```

## pom.xml

```

<project xmlns="http://maven.apache.org/POM/4.0.0"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
        https://maven.apache.org/xsd/maven-4.0.0.xsd">
    <modelVersion>4.0.0</modelVersion>
    <groupId>com.example</groupId>
    <artifactId>MockitoVerifyDemo</artifactId>
    <version>1.0-SNAPSHOT</version>
    <properties>
        <maven.compiler.source>17</maven.compiler.source>
        <maven.compiler.target>17</maven.compiler.target>
    </properties>
    <dependencies>
        <dependency>
            <groupId>org.junit.jupiter</groupId>
            <artifactId>junit-jupiter</artifactId>
            <version>5.10.2</version>
            <scope>test</scope>
        </dependency>
        <dependency>
            <groupId>org.mockito</groupId>
            <artifactId>mockito-core</artifactId>
            <version>5.12.0</version>
            <scope>test</scope>
        </dependency>
    </dependencies>
    <build>
        <plugins>
            <plugin>
                <groupId>org.apache.maven.plugins</groupId>
                <artifactId>maven-surefire-plugin</artifactId>
                <version>3.2.5</version>
            </plugin>
        </plugins>
    </build>
</project>

```



## 6. Logging Error Messages and Warning Levels

The full Project Directory is as follows:

SLF4JLoggingDemo

```
├── .idea
├── src
│   ├── main
│   │   ├── java
│   │   ├── com
│   │   ├── example
│   │   ├── LoggingExample.java
│   │   └── resources
│   └── test
├── target
├── .gitignore
└── pom.xml
```

### Codes:

#### LoggingExample.java

```
package com.example;
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
public class LoggingExample {
    private static final Logger logger =
        LoggerFactory.getLogger(LoggingExample.class);
    public static void main(String[] args) {
        logger.error("This is an error message");
        logger.warn("This is a warning message");
    }
}
```

#### pom.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
        http://maven.apache.org/xsd/maven-4.0.0.xsd">
    <modelVersion>4.0.0</modelVersion>
    <groupId>com.example</groupId>
    <artifactId>LoggingExample</artifactId>
    <version>1.0-SNAPSHOT</version>
    <properties>
        <maven.compiler.source>23</maven.compiler.source>
        <maven.compiler.target>23</maven.compiler.target>
        <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
    </properties>
    <dependencies>
        <!-- SLF4J API -->
        <dependency>
            <groupId>org.slf4j</groupId>
            <artifactId>slf4j-api</artifactId>
            <version>2.0.17</version>
        </dependency>
        <!-- Logback Implementation -->
```

```
    <dependency>
      <groupId>ch.qos.logback</groupId>
      <artifactId>logback-classic</artifactId>
      <version>1.5.18</version>
    </dependency>
  </dependencies>
</project>
```